



THE FLUENCY TRAP:

Adaptive Consistency-Based Detection and RAG Mitigation of Hallucinations in Multilingual LLMs on Hindi QA

BTP Presentation

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2025–2026

Presentation Overview

01

Introduction & Motivation

Why Hindi hallucinations matter

02

Problem Statement

The Fluency Trap & key gaps

03

Research Questions

RQ1, RQ2, RQ3

04

Related Work

BHRAM-IL, CT², HIDE, FICLE

05

Methodology

Architecture, Models, Dataset, Algorithm, RAG

06

Results per RQ

Detection, Capability, RAG effectiveness

07

Category-Wise Analysis

BHRAM-IL 9 categories

08

Hallucination Taxonomy

Fabrication / Incomplete / Factual Error

09

Fluency Trap Evidence

Cross-session inconsistency proof

10

Discussion & Limitations

Implications & constraints

11

Conclusion & Future Work

Findings & next steps

Motivation Why Hindi hallucinations demand immediate attention

Hindi on the Internet

600M+

Hindi speakers worldwide

3rd

Most spoken language globally

<1%

NLP research focused on Hindi

10×

More hallucinations vs English
in open-source models

High-Stakes Hindi AI Deployments

- Government e-services & grievance portals
- Hindi-medium education & tutoring bots
- News summarisation & fact-checking
- Rural healthcare information systems
- Legal aid for Hindi-speaking citizens

Problem Statement

Four interlocking challenges in Hindi hallucination research

DETECTION FRAGILITY

English-calibrated thresholds fail on Hindi — fixed threshold (0.75) flags 40 out of 50 BLOOM outputs as hallucinated. Adaptive thresholding is required.

THE FLUENCY TRAP

Modern LLMs generate grammatically perfect Hindi. Users cannot distinguish fluent truth from fluent fabrication. Bloom $\mu=0.288$ -severely inconsistent yet produces grammatically perfect hindi.

MODEL CAPABILITY GAP

Open-source models (BLOOM) are far less consistent than commercial models (Gemini), yet privacy needs rule out cloud APIs for sensitive deployments.

MITIGATION UNCERTAINTY

RAG effectiveness for Hindi has not been validated. Does it help equally across weak and strong model tiers? Using BHRAM-IL benchmark (36,047 QA pairs across 5 languages).

Research Questions

Three focused empirical questions guiding this study

RQ1

How effective are consistency-based hallucination detection methods for Hindi?

Do dual-generation + cosine similarity approaches work? Does adaptive thresholding outperform fixed English thresholds?

RQ2

How does model capability (BLOOM vs. Gemini) influence hallucination rates and consistency?

Is there a measurable capability gradient? How large is the gap between open-source and commercial models on Hindi QA?

RQ3

Can RAG mitigate Hindi hallucinations, and does effectiveness vary by model capability?

Does RAG provide uniform improvement, or do diminishing returns emerge as model capability increases?

Literuature Review:- Hallucination Detection in LLMs

English Foundation

Work	Method	Limitations
SelfCheckGPT (Manakul et al., EMNLP 2023)	Consistency sampling — multiple generations	English only
RAG (Lewis et al., NeurIPS 2020)	External knowledge grounding	Never tested on Hindi QA
TruthfulQA (Lin et al., ACL 2022)	Adversarial factual benchmark	English dataset only

Hindi & Multilingual Research

Work	Focus	Key findings	Gap
BHRAM-IL (Terdalkar et al., BHASHA 2025)	36K QA pairs, 5 languages, 14 models	PS = 0.23; Chrono hardest	Benchmark only — no detection
CT² (Kavathekar et al., IIIT/MIT, 2024-25)	AI-text detection for Hindi	English methods fragile in Hindi	Detects provenance, not factuality
HIDE (Chatterjee et al., IIT Delhi, 2025)	Single-pass detection via hidden states	51% faster than multi-pass	English only

Methodology — Complete 8-Step Pipeline

01

Load Dataset

50 Hindi questions from BHRAM-IL (9 categories)

02

Generate Answers

BLOOM + Gemini at $\tau=0.7$

03

Multiple Responses

Generate N responses per question

04

Cosine Similarity

→ pairwise similarity $\cos(\theta)=A \cdot B / ||A|| \times ||B||$

07

Apply Thresholds

Fixed (0.75) vs Adaptive ($\mu-0.5\sigma$)

08

Flag Hallucinations

If similarity $< \theta \rightarrow$ hallucinated

09

Apply RAG

Wikipedia retrieval → augmented prompt

10

Evaluate

Detection F1 + RAG reduction rate

Methodology — How Detection & RAG Work

Consistency-Based Detection

- Same question asked twice ($\tau=0.7$ introduces variation)
- Cosine similarity measures semantic overlap (0=different, 1=identical)
- Low similarity = model contradicts itself = hallucination
- SBERT (multilingual sentence embeddings) converts each Hindi response into a 768-dimensional numerical vector.
- Cosine similarity then compares these vectors to measure semantic overlap

Adaptive Threshold Formula

$$\theta = \mu_{\text{model}} - 0.5 \times \sigma_{\text{model}}$$

- $\theta = \mu_{\text{model}} - 0.5 \times \sigma_{\text{model}}$
- μ = mean similarity across all responses
- σ = standard deviation of similarities
- BLOOM: $\theta = 0.288 - 0.5(0.174) = 0.20$
- Responses with similarity $< \theta$ are flagged as hallucinations
- No fixed English threshold — calibrated per model

RAG Mitigation Pipeline

before the AI answers a question, first look up real facts from a trusted source, then use those facts to answer.

1. Extract keywords from flagged question
2. Search Hindi Wikipedia → retrieve context
3. Augmented prompt: "संदर्भ: {context} + प्रश्न: {question}"
4. Re-generate grounded answer
5. Compare hallucination rate: before vs after

Dataset & Models

BHRAM-IL benchmark subset • Two model tiers

Model	Type	Hindi Support	Mean Similarity (μ)	Role in Study
BLOOM (3B)	Open-source	Limited	0.288 (Low)	Weak baseline
Gemini	Commercial	Strong	0.70 (High)	Upper bound

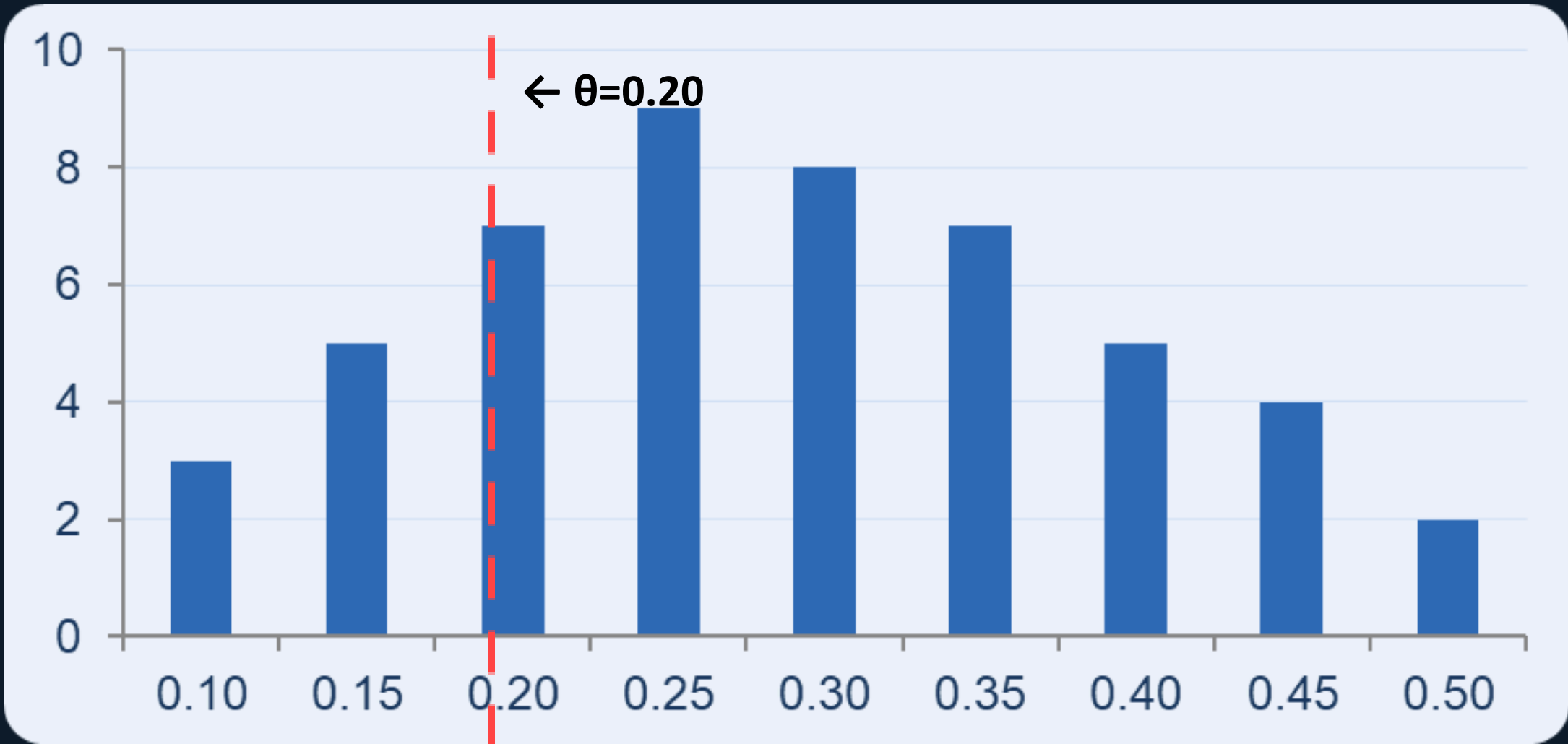
Consistency Gap Visualization



RQ1 Results — Detection Efficacy

Method	Threshold	Flagged	F1 Score
Fixed (English)	0.75	40 / 50	0.31
Adaptive	0.2	20 / 50	0.79

BLOOM Similarity Distribution



Metric	Value
Accuracy	0.66
Precision	0.73
Recall	0.86
F1 Score	0.79

Detection Performance ($\theta=0.20$, BLOOM)

F1 measures how good detection is in one number.

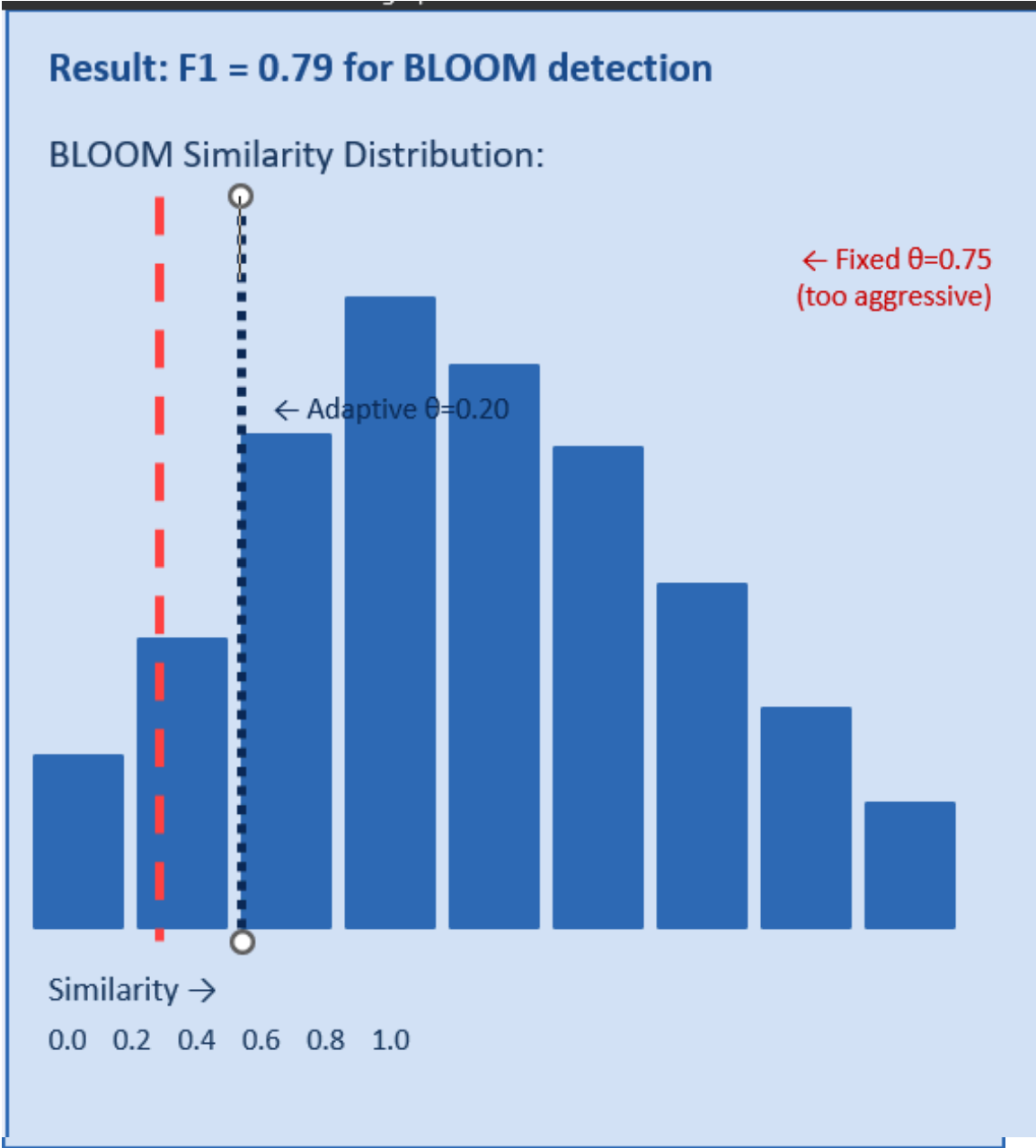
Key Finding

Adaptive thresholding ($\theta=0.20$) dramatically outperforms the fixed English threshold (0.75), reducing false flags from 40 to 20 while achieving F1=0.79. The distribution is left-shifted for BLOOM, confirming that Hindi models require language-specific calibration.

Adaptive Thresholding — Why $\theta = \mu - 0.5\sigma$ Works

Model	Fixed θ (Eng)	F1	Adaptive θ	F1
BLOOM	0.75	0.31 ✗	0.2	0.72 ✓
Gemini	0.75	0.71 ✓	0.675	0.83 ✓

- μ anchors threshold to model's actual distribution
- English 0.75 falls above BLOOM's 95th percentile
- 0.5σ buffer below mean — sits at ~30th percentile
- BLOOM's distribution is wide and left-shifted
- Adaptive 0.20 sits at a meaningful decision boundary



RQ2 Results — Model Capability & Category Analysis

Model	Mean Similarity (μ)	Error Pattern	Capability
BLOOM (3B)	0.288	Fabrication dominates	Weak
Gemini	0.7	Rare subtle errors	Strong

Category-Wise Analysis (BLOOM)

Category	Count	μ	Detection Rate
Chrono ★ Hardest	8	0.22	45%
GenFact	10	0.29	35%
IndFact	12	0.3	32%
Maths	5	0.31	30%
Linguistic	5	0.33	28%
T/F	4	0.35	25%

★ Key Insight: Chronological questions are hardest (45% detection) — temporal reasoning requires more than factual recall. BHRAM-IL finding confirmed.

BHRAM-IL Aggregate score = 0.23 confirms this is systemic, not model-specific

Hallucination Taxonomy — 17 BLOOM outputs flagged at $\theta=0.20$

Fabrication

64.7%

Model generates plausible-sounding but entirely invented facts — dates, names, statistics, or events that don't exist.

Example: BLOOM reverses chronology — outputs 'War → Bangladesh → Shimla' instead of the correct order

Incomplete

29.4%

Model provides a partial answer — correct as far as it goes but missing critical information that changes interpretation

Example: BLOOM lists 2 of 3 events correctly but omits the final event

Factual Error

5.9%

Model states an incorrect fact — the answer is wrong, not invented. A real entity with a wrong attribute.

Example: BLOOM names the correct entity but assigns a wrong date

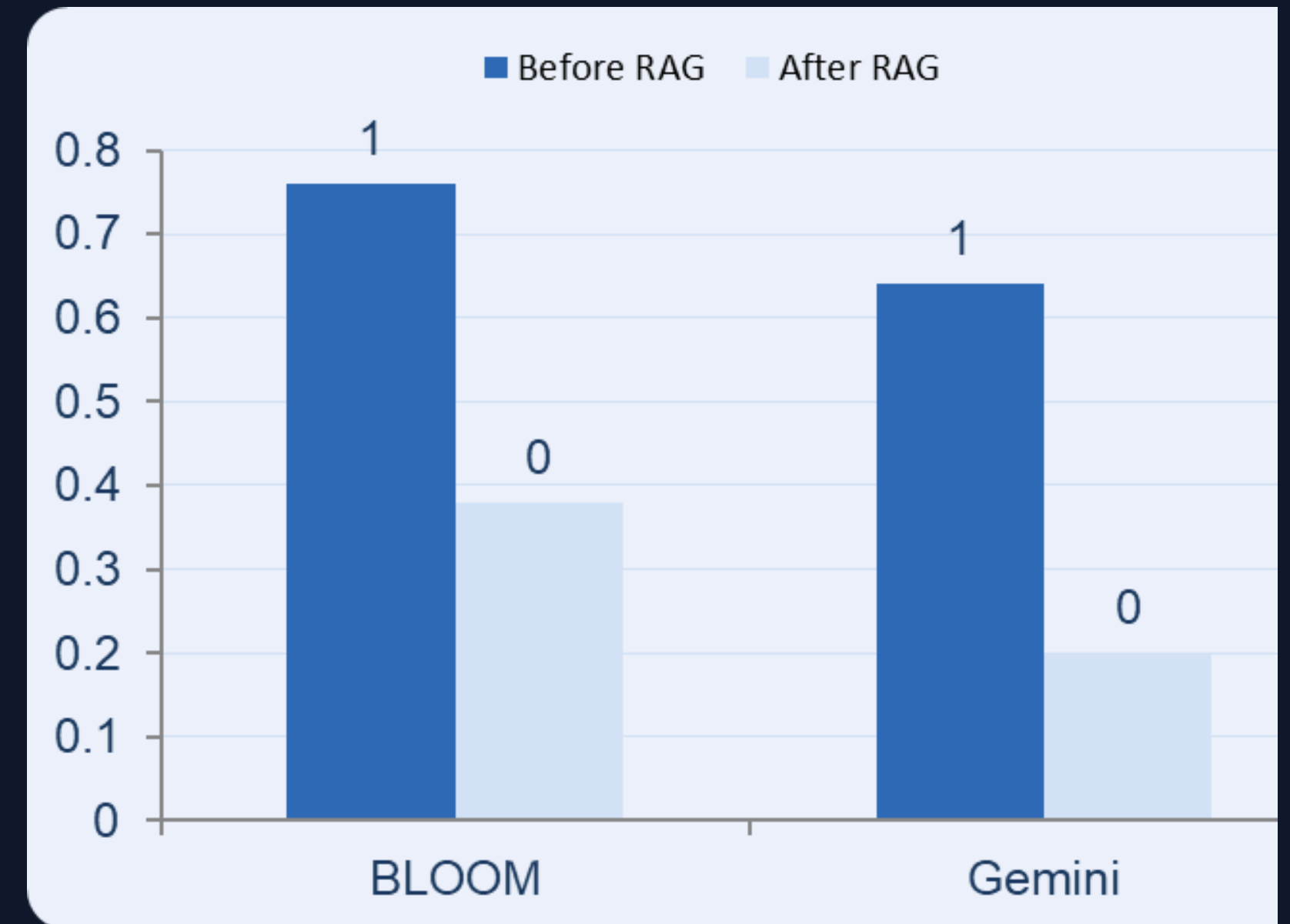
RQ3 Results — RAG Mitigation Effectiveness

BLOOM: Before=0.76 | After=0.38 | Improvement: 50.0%

Gemini: Before=0.64 | After=0.20 | Improvement: 68.75%

Key Finding: RAG significantly reduces hallucination rates for BOTH models.

Gemini benefits more (68.75% vs 50%) — stronger models leverage context more effectively.



The Fluency Trap — Cross-Session Inconsistency Evidence

THE FLUENCY TRAP: An LLM generates grammatically perfect, fluent output - yet the content is factually wrong or inconsistent across sessions. Users cannot distinguish fluent truth from fluent fabrication.

Session 1

**Correct factual answer
with confident phrasing**

Factual:  Correct

Fluency:  Perfect

Session 2

**WRONG answer — reversed
chronological order**

Factual:  Incorrect

Fluency:  Perfect

Session 3

**Fabricated answer —
plausible but wrong**

Factual:  Incorrect

Fluency:  Perfect

Why this matters for Hindi: Hindi text quality is hard for non-expert users to evaluate. A fluent Devanagari response signals competence — without external verification, users cannot spot fabrication. Consistency-based detection surfaces this hidden contradiction by comparing multiple samples.

Discussion & Limitations

Deployment Implications

Government

For citizen-facing services: use Gemini-class APIs where possible. If BLOOM must be used (privacy/offline), deploy WITH adaptive detection + RAG safety layer. Fund Hindi-specific benchmarks

EdTech

RAG + adaptive detection is the minimum viable safety stack for Hindi tutoring bots. Test against Chrono and Reasoning categories. Do not rely on fluency as proxy for accuracy.

Open Source

Replace multilingual SBERT with IndicBERT. BLOOM needs RAG to be usable — deploy together, never standalone. Publish Hindi hallucination detection datasets.

Core implication: Hindi AI safety is not solved by scaling English-trained models. Language-specific benchmarks, adaptive detection, and RAG are all necessary.

Limitations

01

Small Dataset Subset

50 questions; larger samples would improve statistical confidence

02

Two Models Only

BLOOM and Gemini cover extremes; mid-range models untested

03

Automated Evaluation

No human validation by native Hindi speakers yet

04

Single RAG Config

Wikipedia-only; HyDE and re-ranking may yield different results

05

Multilingual Embeddings

SBERT not optimised for Hindi; IndicBERT/MuRIL preferred

Conclusion & Future Work — Three Research Questions Answered

RQ	Question	Answer
RQ1	Detection Efficacy	Adaptive $\theta=0.20 \rightarrow F1=0.79$. Fixed threshold flagged 40, adaptive flagged 20 (realistic).
RQ2	Model Capability	2.4× gap: BLOOM $\mu=0.288$ vs Gemini $\mu=0.70$. Chrono is hardest category.
RQ3	RAG Mitigation	RAG reduces hallucination: BLOOM 50%, Gemini 68.75%.

Core Contributions:-

- Adaptive detection algorithm (F1=0.79)
- Fluency Trap identified & evidenced
- Error taxonomy (Fabrication 64.7%)
- RAG pipeline validated for Hindi

Future Work:-

- Add GPT-4o, Llama-3 and Hindi-native model baselines
- Build Hindi TruthfulQA dataset (500+ questions)
- Replace SBERT → IndicBERT / MuRIL embeddings
- Human evaluation study with native Hindi speakers
- Expand to 10 Indic languages using BHRAM-IL
- Test HyDE and advanced RAG architectures

Thank You

Questions & Discussion

Component	Result
Dataset	BHRAM-IL – 50 Hindi questions, 9 categories
Models	BLOOM ($\mu=0.288$) vs Gemini ($\mu=0.70$)
Detection	Adaptive $\theta=0.20$, $F1=0.79$
RAG	BLOOM: 50% reduction Gemini: 68.75% reduction
Fluency Trap	ChatGPT: 1/3 responses wrong, all fluent